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Contents

Contents i

1 We gotta start somewhere 1

1.1 Minimizing Energy 1

Bibliography 7

Chapter 1

We gotta start somewhere

Given a finite connected graph K and it vertices V_m the for a function $u : V_m \to \mathbb{R}$ we define the energy to be

$$E_K(u) = \sum_{x \sim y} \big(u(x) - u(y)\big)^2$$

1.1 Minimizing Energy

Given some graph K and a function $u^{(m)} : V_m \to \mathbb{R}$, let's say we want to extend this function to V_{m+1}. This leads us to defining an extension in the following way

Definition 1.1: Let K be a graph and $u : V_m \to \mathbb{R}$, then an *extension of* u is a function $u' : V_{m+n} \to \mathbb{R}$ where $u' \mid_{V_m} = u$.

It is obvious that $E_{m+n}(u') \geq E_m(u)$. Further one quickly notices that there must be a function that minimizes the energy of a given graph. We denote such a function with $\tilde{u}$ and call it a *harmonic extension*.

Definition 1.2: Let $\tilde{u} : V_{m+n} \to \mathbb{R}$ with $\mid_{V_m} = u$ is called a *harmonic extension* if $E_{m+n}(\tilde{u}) \leq E_{m+n}(u')$ for all $u' : V_{m+n} \to \mathbb{R}$ with $u' \mid_{V_m} = u$.

In some cases we have that $E_{m+n}(\tilde{u}) = rE_m(u)$ for some $r \in (0, 1)$. But instead of normalizing the values on E_m we will normalize E_{m+n}, and getting

$$r^{-1}E_{m+n}(\tilde{u}) = E_m(u), \quad \text{and} \quad r^{-1}E_{m+n}(u') \geq E_m(u)$$

for every extension u' of u. From the equality above we can define the following

Definition 1.3: The *renormalized graph energies* are defined as

$$\mathcal{E}_m(u) = r^{-m}E_m(u)$$

Remark 1.4: The sequence $\big\{\mathcal{E}_m(u)\big\}$ is a non-decreasing sequence. Something with it being constant if u is linear and how is u defined? In the sequence $\big\{\mathcal{E}_m(u)\big\}$ if we have u defined on V_m what is $\mathcal{E}_{m+1}(u)$? Any extension or the harmonic extension?

Now that we have 1.3, we can define what it means to be a harmonic function.

Definition 1.5: u is a *harmonic function* if u minimizes $\mathcal{E}_m(u)$ for all m, given boundary values V_0.

Minimizing Energy on the Vicsek set

I the previous section we have talked about a harmonic function, but without every constructing one. We will therefore in this next section show how to create an harmonic function for the Vicsek set for any start boundary values.

Given $V_0 = \{q_0, q_1, q_2, q_3, q_4\}$ and $u : V_0 \to \mathbb{R}$ with $u(q_i) = \alpha_i$ for $i = 1, 2, \ldots, 5$. Then we have that

$$E_0(u) = \big(u(q_0) - u(q_1)\big)^2 + \big(u(q_0) - u(q_2)\big)^2 + \big(u(q_0) - u(q_3)\big)^2 + \big(u(q_0) - u(q_4)\big)^2 .$$

We are going to exploit symmetry of the Vicsek set. Since all four "limbs" of the Vicsek set are identical we will only look at one of the limbs since the other limbs will be the same computations. So if we want to minimize $E_1(u')$, we see (for one limb) that

$$E_1(u') = \big(u'(q_1) - u'(p_0)\big)^2 + \big(u'(p_0) - u'(p_2)\big)^2 + \big(u'(p_0) - u'(p_3)\big)^2 + \big(u'(p_0) - u'(p_4)\big)^2 + \big(u'(p_3) - u'(q_0)\big)^2$$

We are then going to optimize with regards to $u'(p_i)$ for $i = 1, 2, 3$. Taking the derivative of the right side and solving for 0 we get the following

$$\begin{aligned} 4u'(p_0) &= u'(q_1) + u'(p_2) + u'(p_3) + u'(p_4) \\ 2u'(p_3) &= u'(q_0) + u'(p_0) \\ u'(p_2) &= u'(p_0) \\ u'(p_4) &= u'(p_0). \end{aligned}$$

Solving these equations we get

$$\begin{aligned} u'(p_0) &= \frac{2}{3}u'(q_1) + \frac{1}{3}u'(q_0) \\ u'(p_2) &= \frac{2}{3}u'(q_1) + \frac{1}{3}u'(q_0) \\ u'(p_4) &= \frac{2}{3}u'(q_1) + \frac{1}{3}u'(q_0) \\ u'(p_3) &= \frac{2}{3}u'(q_0) + \frac{1}{3}u'(q_1). \end{aligned}$$

So what we have found is that to minimize the energy of the Vicsek set, then the energy of the next point is going to be 2/3 the energy from the closest point and 1/3 the energy from the next closest point.

Lemma 1.6: Let $v : K \to \mathbb{R}$ be a harmonic function on the Vicsek set. If $x \underset{m}{\sim} y_i$ for and $v(x) = v(y_i) = 0$ for all neighbours y then $v(\omega) = 0$ for all $\omega \in \overline{B_x}$ where $B_x = \left(\bigcup_{|w|>0} F_w x\right) \cup \left(\bigcup_i \{y_i\}\right)$

Proof. It is clear that $v(\omega) = 0$ for any $\omega \in B_x$, since the energy of the points $F_w x$ with $|w| > 0$ depends on the energy of x and its neighbours.

So we need to check the points in $\overline{B_x} \backslash B_x$. Now since B_x is dense in $\overline{B_x}$, then for any $\omega \in \overline{B_x}$ we have that there exists a sequence $x_n \in B_x$ such that $x_n \to \omega$. By the fact that v is continuos on K it is continuos on $\overline{B_x}$, in which case $\big|v(x_n) - v(\omega)\big| \to 0$ as $n \to \infty$, but $v(x_n) = 0$ for all x_n, so $v(\omega) = 0$ as desired. ☐

Lemma 1.7: Set $F_m = \operatorname{supp} \psi_x^{(m)}$, then

$$F_0 \supset F_1 \supset F_2 \supset F_3 \supset \ldots$$

Further F_m is contained in the open ball $B(x, \epsilon_m)$ where $\epsilon_m \to 0$ as $m \to \infty$.

Proof. If we set $\Omega_m = \left\{ z \in K : \text{ if } z \underset{m}{\sim} y \text{ then } \psi_x^{(m)}(z) = \psi_x^{(m)}(y) = 0 \right\}$ then $F_m = K \backslash \left(\bigcup_{\alpha \in \Omega_m}\right) \overline{B_\alpha}$. It is clear that $\Omega_m \subset \Omega_{m+1}$ and that there are new points in Ω_{m+1} that are not just a point between some already existing points $x_1, x_2 \in \Omega_m$. We therefore have from the way that F_m is constructed that $F_m \supset F_{m+1}$.

The containment of F_m in $B(x, \epsilon_m)$ follows from lemma 1.6 and that $F_m \supset F_{m+1}$. ☐

Theorem 1.8: Let $u \in \operatorname{dom} \Delta_\mu$. Then the pointwise formula

$$\Delta_\mu u(x) = \lim_{m \to \infty} r^m \left(\int_K \psi_x^{(m)} \, \mathrm{d}\, \mu \right)^{-1} \delta_m u(x)$$

holds with the limit uniform across $V_* \backslash V_0$. Conversely, suppose u is a continuos function and the right side of the limit converges uniformly to a continuos function on $V_* \backslash V_0$. Then $u \in \operatorname{dom} \Delta_\mu$ and the equation holds.

Proof. Suppose $u \in \operatorname{dom} \Delta_\mu$ and $\Delta_\mu u = f$, then by [INSERT REF] we have that

$$r^{-m} \left(\int_K \psi_x^{(m)} \,\mathrm{d}\,\mu \right)^{-1} \Delta_m u(x) = \frac{\int_K f \psi_x^{(m)} \,\mathrm{d}\,\mu}{\int_K \psi_x^{(m)} \,\mathrm{d}\,\mu}$$

for any $x \in V_m \backslash V_0$. Uniform converges of the right side is because of continuity of $f(x)$, so let $\epsilon > 0$ be given, then by lemma 1.7 we can choose m large enough so for all $x, y \in F_m$ that $\left| f(x) - f(y) \right| < \epsilon$, then

$$\begin{aligned}
\left| \frac{\int_K f \psi_x^{(m)} \,\mathrm{d}\,\mu}{\int_K \psi_x^{(m)} \,\mathrm{d}\,\mu} - f(x) \right| &= \left| \frac{\int_K f \psi_x^{(m)} \,\mathrm{d}\,\mu}{\int_K \psi_x^{(m)} \,\mathrm{d}\,\mu} - \frac{\int_K f(x) \psi_x^{(m)} \,\mathrm{d}\,\mu}{\int_K \psi_x^{(m)} \,\mathrm{d}\,\mu} \right| \\
&= \left| \frac{\int_K \big(f(y) - f(x) \big) \psi_x^{(m)} \,\mathrm{d}\,\mu}{\int_K \psi_x^{(m)} \,\mathrm{d}\,\mu} \right| \\
&\leq \frac{\int_{F_m} \left| f(y) - f(x) \right| \mathrm{d}\,\mu}{\int_K \psi_x^{(m)} \,\mathrm{d}\,\mu} \\
&\leq \frac{\epsilon \int_{F_m} \mathrm{d}\,\mu}{\int_K \psi_x^{(m)} \,\mathrm{d}\,\mu} \\
&= \epsilon
\end{aligned}$$

in which case

$$\lim_{m \to \infty} r^{-m} \left(\int_K \psi_x^{(m)} \,\mathrm{d}\,\mu \right)^{-1} \Delta_m u(x) = \lim_{m \to \infty} \frac{\int_K f \psi_x^{(m)} \,\mathrm{d}\,\mu}{\int_K \psi_x^{(m)} \,\mathrm{d}\,\mu} = f(x) = \Delta_\mu u(x)$$

Moving on to the second half of the proof. Suppose that $r^m \left(\int_K \psi_x^{(m)} \,\mathrm{d}\,\mu \right)^{-1} \delta_m u(x)$ converges uniformly to a continuos function f. Now for any $v \in \operatorname{dom}_0 \mathcal{E}$ we have

$$\mathcal{E}_m(u, v) = r^{-m} \sum_{x \underset{m}{\sim} y} \big(u(y) - u(x) \big) \big(v(y) - v(x) \big) .$$

If we collect all the terms that contain the factor $v(x)$ for a fixed $x \in V_m \backslash V_0$ then each $v(x)$ will be of the form $-v(x) \big(u(y) - u(x) \big)$ for each $y \underset{m}{\sim} x$. In which case we get

$$\begin{aligned}
\mathcal{E}_m(u, v) &= -r^{-m} \sum_{x \in V_m \backslash v_0} v(x) \left(\sum_{x \underset{m}{\sim} y} \big(u(y) - u(x) \big) \right) \\
&= - \sum_{x \in V_m \backslash V_0} v(x) r^{-m} \Delta_m u(x)
\end{aligned}$$

While it seems we lose many terms by removing $v(y)$, but they are instead contained in $\sum_{x \underset{m}{\sim} y} \big(u(y) - u(x) \big)$. Note that the sum is "reset" for every x so we count both the relation $x \sim y$ and $y \sim x$.

Now if we define $f_m(x) = r^{-m} \left(\int_K \psi_x^{(m)} \,\mathrm{d}\,\mu \right)^{-1} \Delta_m u(x)$ then $f_m \to f$ uniformly by assumption and we can rewrite the above as

$$\begin{aligned}
\mathcal{E}_m(u, v) &= - \sum_{x \in V_m \backslash V_0} v(x) \left(\int_K \psi_x^{(m)} \,\mathrm{d}\,\mu \right) r^{-m} \left(\int_K \psi_x^{(m)} \,\mathrm{d}\,\mu \right)^{-1} \Delta_m u(x) \\
&= - \int_K \left(\sum_{x \in V_m \backslash V_0} v(x) f_m(x) \psi_x^{(m)} \right) \mathrm{d}\,\mu.
\end{aligned}$$

Following a quite lengthy argument, we have that

$$\mathcal{E}(u,v) = \lim_{m\to\infty} \mathcal{E}_m(u,v) = \lim_{m\to\infty} -\int_K \left(\sum_{x\in V_m\setminus V_0} v(x) f_m(x) \psi_x^{(m)} \right) \mathrm{d}\,\mu. = -\int_K v(y)f(y)\,\mathrm{d}\,\mu.$$

$$\begin{aligned}\left| \sum_{x\in V_m\setminus V_0} v(x) f_m(x) \psi_x^{(m)}(y) - v(y)f(y) \right| \\ \leq \left| \sum_{x\in V_m\setminus V_0} \big(v(x)f_m(x) - v(y)f(y)\big)\, \psi_x^{(m)}(y) \right| + \left| \sum_{x\in V_m\setminus V_0} v(y)f(y)\psi_x^{(m)}(y) - v(y)f(y) \right|\end{aligned}$$

Looking at each term individually we see that

$$\begin{aligned}\left| \sum_{x\in V_m\setminus V_0} \big(v(x)f_m(x) - v(y)f(y)\big)\, \psi_x^{(m)} \right| \\ \leq \sum_{x\in V_m\setminus V_0} \Big(\big|v(x)f_m(x) - v(y)f_m(y)\big| + \big|v(y)f_m(y) - v(y)f(y)\big| \Big)\, \psi_x^{(m)}(y)\end{aligned}$$

now again we will look at each term starting with $\big|v(y)f_m(y) - v(y)f(y)\big|$. Here

$$\big|v(y)f_m(y) - v(y)f(y)\big| = \big|v(y)\big|\big|f_m(y) - f(y)\big|$$

and since v is continuos on K, which is compact, there is a constant C_1 so $\big|v(y)\big| \leq C_1$ for all $x \in K$. Since f_m converges uniformly to f we can choose m_1 such that $\big|f_m(y) - f(y)\big| \leq \frac{\epsilon}{C_1}$ and the convergence of that term follows.

Next is $\big|v(x)f_m(x) - v(y)f_m(y)\big|$, here we find

$$\big|v(x)f_m(x) - v(y)f_m(y)\big| \leq \big|v(x)\big|\big|f_m(x) - f(y)\big| + \big|f_m(y)\big|\big|v(x) - v(y)\big|$$

First $\big|f_m(y)\big|\big|v(x) - v(y)\big|$, here $\big|f_m(y)\big| \leq C_2$ is uniformly bounded by some C_2 why?. By lemma 1.7 and continuity of v we can choose m_2 so $\big|v(x) - v(y)\big| \leq \frac{\epsilon}{C_2}$ for all $x \in F_{m_2}$. The reason we can choose m_2 is because $\psi_x^{(m)}$ is multiplied onto all the terms so we just restrict v to F_{m_2}.

Second $\big|v(x)\big|\big|f_m(x) - f(y)\big|$, again $\big|v(x)\big| \leq C_1$, but

$$\big|f_m(x) - f(y)\big| \leq \big|f_m(x) - f(x)\big| + \big|f(x) - f_m(y)\big| \leq \big|f_m(x) - f(x)\big| + \big|f(x) - f(y)\big| + \big|f(y) - f_m(y)\big|$$

First choose m_3 so $\big|f_{m_3}(x) - f(x)\big| \leq \frac{\epsilon}{3C_1}$, next by continuity of f choose m_4 so $\big|f(x) - f(y)\big| \leq \frac{\epsilon}{3C_1}$ for all $x \in F_{m_4}$, lastly choose m_5 so $\big|f(y) - f_{m_5}(y)\big| \leq \frac{\epsilon}{3C_1}$.

So we now have for $m \geq \max\{m_1, m_2, \ldots, m_5\}$ that

$$\left| \sum_{x\in V_m\setminus V_0} \big(v(x)f_m(x) - v(y)f(y)\big)\, \psi_x^{(m)} \right| \leq \epsilon \sum_{x\in V_m\setminus V_0} \psi_x^{(m)} \leq \epsilon \sum_{x\in V_m\setminus V_0} 1 \leq \epsilon.$$

We are now going to look at $\left|\sum_{x\in V_m\setminus V_0} v(y)f(y)\psi_x^{(m)}(y) - v(y)f(y)\right|$ here

$$\left| \sum_{x\in V_m\setminus V_0} v(y)f(y)\psi_x^{(m)}(y) - v(y)f(y) \right| = \big|v(y)f(y)\big| \left| \sum_{x\in V_m\setminus V_0} \psi_x^{(m)}(y) - 1 \right|$$

Now to show $\left|\sum_{x\in V_m\setminus V_0} \psi_x^{(m)}(y) - 1\right| = 0$ as $m \to \infty$, we start by looking at $y \in V^*$. Choose m_6 so $y \in V_{m_6}$.

Next assume $y \in K$, then since V^* is dense in K we can choose a sequence $(x_n)_{n=1}$ where $x_n \to y$ as $n \to \infty$, and since $\psi_x^{(m)}$ is continuos on K then $\psi_x^{(m)}(x_n) \to \psi_x^{(m)}(y)$ as $n \to \infty$. This gives us the following

$$\left| \sum_{x \in V_m \setminus V_0} \psi_x^{(m)}(y) - 1 \right| \leq \left| \sum_{x \in V_m \setminus V_0} \psi_x^{(m)}(y) - \sum_{x \in V_m \setminus V_0} \psi_x^{(m)}(x_n) \right| + \left| \sum_{x \in V_m \setminus V_0} \psi_x^{(m)}(x_n) - 1 \right|$$

Now choose an n_1 so $\left| \psi_x^{(m)}(x_{n_1}) - \psi_x^{(m)}(y) \right| \leq \epsilon$ and an m_7 so $x_{n_1} \in V_{m_7}$. Both can then be bounded by $m \geq \max\{n_1, m_7\}$ $\square$

Bibliography

[Str06] Robert S. Strichartz. *Differential Equations on Fractals: A Tutorial.* Princeton University Press, 2006.